# What is in Kinetic_Modeling, and how it got here

The repository on 20 October 2026: its four generations of KrF models, what each one fixed, and where everything lives

**Bottom line.** Kinetic_Modeling models the plasma kinetics of KrF (and ArF) excimer lasers. It has had four generations, each begun because the one before could not be trusted: a basic ODE recreation of Chow's 1978 laser; phantom, which took its electron rates from LoKI-MC because a Maxwellian cannot follow inverse-bremsstrahlung heating, and which failed on bugs and on calibration it did not declare; SPECK, a clean rebuild (the REBUILD chemistry engine beside a 3-D particle code); and LIGHT, a second and much smaller rebuild, cheaper to run than SPECK and now the most accurate 0-D model here (the held-out KrF decays at $\times 0.95$ against REBUILD's $\times 1.37$, the Tahoma gain at $\times 0.93$ against $\times 0.58$).

Use LIGHT for $Kr/F_2$ in 0-D. REBUILD is still the only model here for an argon or neon buffer, a real drive, a cavity or a 3-D grid.

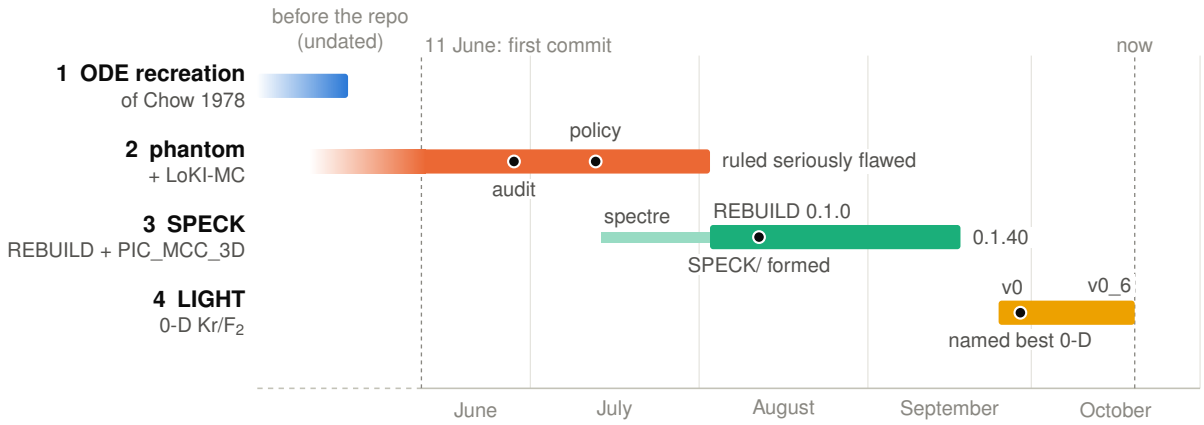



Figure 1: The four generations in time. A bar that fades to the left began before the repository's first commit (11 June 2026), so its start is not recorded. The dots are milestones named in Section 2.

## 1 What the repository is

A Python research repository for the plasma kinetics of KrF and ArF excimer lasers: how electrons excite and ionize the gas, how the KrF* excimer forms and is destroyed, and how much of the input power leaves as 248-nm light. One question runs through all of it: can reheating the electrons, with a $CO_2$ laser (inverse bremsstrahlung) or a microwave, make the laser more efficient? Section 4 gives each generation's answer.

Since 13 July 2026 every model here runs under one rule, the predictive-only policy at the top of `PLAN_physical_validity.txt`: each rate comes from theory or from a measurement of that process, and a benchmark scores a result after the fact but never sets a number.

## 2 History, in four generations

### 1. Chow, recreated with an ODE solver (before June 2026)

The work began as a 0-D recreation of Chow, Hohla and Rebentrost's 1978 discharge-pumped KrF laser (the Mephisto device: 1.5 bar of He:Kr:$F_2$ = 90:9.5:0.5 and a 50-ns discharge; `docs/KrF_Chow.pdf`).

`legacy_code.py` integrates 11 variables and `KrF_ODE_original.py` 13 (electrons, their temperature, photons and the excited and ionized species) with scipy's `solve_ivp`, and every electron-impact rate is an average over a Maxwellian (`Kmaxwell`). Both are kept unedited in `KrF_and_LoKI/KrF Code/og-pyfiles/`, as the reference that later refactoring was checked against.

## 2. phantom, with LoKI-MC (to 3 August 2026)

The ODE model grew, version by version (v12 to v35, then v2.1.1, all before the repository's first commit), into `KrF_phantom11.py`: one 9,798-line file, later split into the 26-species `phantom11` package and the GUIs at the root. A Maxwellian cannot follow electrons that inverse bremsstrahlung is heating, since the heating drives the distribution away from a Maxwellian (the Langdon effect), so from v26 the electron rates came from LoKI-MC's Boltzmann solutions; LoKI-MC itself is kept in `LoKI-MC/`. phantom failed on bugs and on calibration it did not declare:

- The audit of 28 June 2026 (`docs/PHANTOM_AUDIT.md`) found its matches to Chow, Ong and Chen circular. The electron density was capped at the answer ($10^{16}\,\mathrm{cm}^{-3}$), E/N was re-tuned for each benchmark, energy was never balanced, and one LoKI table entry was ten times too small.
- The LoKI rates were evaluated at an electron temperature tuned to Chow's measurement, not the one LoKI computes (`KrF_and_LoKI/docs/two-region-eedf.md`).
- `PLAN_physical_validity.txt`, entries (20) to (74), kept finding defects, among them an $ArF^*$ loss summed as a source (r115) and a physics default, chosen to reproduce the frozen originals, that changes the light output 3.4 times (entry (72)).

On 13 July 2026 the predictive-only policy was written and phantom's benchmark path banned (`old_files/README.md`). On 3 August phantom was ruled "seriously flawed".

## 3. SPECK, rebuilt through REBUILD (August to September 2026)

The first step away was spectre (14 July), a small neon-buffered engine whose changelog opens "Following the failed attempt of phantom". The rebuild plan of 3 August (`SPECK/ORESTES_CLASS_REBUILD.txt`) argued for hardening spectre; the decision was a new package instead, REBUILD (0.1.0 on 4 August, 0.1.40 on 18 September), with about 170 reactions among 35 species, built so that phantom's defects cannot come back:

- its reaction network is data, each row with its source;
- it owes nothing to a frozen reference, so no physics default is set by matching old code;
- its energy ledger is solved with the physics (worst residual $8.2 \times 10^{-6}$ over a pulse);
- a test fails if a tuning knob appears.

It shares no code with phantom, though rate values came across with their citations. On 12 August it was joined under `SPECK/` with PIC_MCC_3D, a 3-D electrostatic particle-in-cell code begun on 3 August that measures the non-Maxwellian rates; the two are coupled by data only. Its limits: it predicts too little gain (Klimek $\times 0.64$, Tahoma $\times 0.58$, Electra ArF $\times 0.75$), and a multi-cell run takes hours to days.

## 4. LIGHT, rebuilt again (25 September 2026 to now)

LIGHT started over from `KrF_model_notes_with_Gary.rtf`: one well-mixed 0-D box of Kr and $F_2$, 24 reaction rows in v0 against REBUILD's 174, a run in under a second. Its default network ties each rate to a measurement made on KrF, and `ML/` fits the KrF(B,C) constants to the decay data. The comparison of 29 September (`comparisons/README.md`) scored it beside REBUILD and Nathaniel's Fluid1D on the same measurements and named it the best 0-D model: the held-out Xu 1993 decays at $\times 0.95$ (REBUILD $\times 1.37$) and the Tahoma e-beam gain at $\times 0.93$ (REBUILD $\times 0.58$). The same comparison calls the Tahoma agreement encouraging, not proof: it moves to $\times 0.86$ with REBUILD's ion-ion B/C

split, which is equally unmeasured, and to $\times 0.46$ with the weaker trimer temperature law. LIGHT is $Kr/F_2$ only by design, with no drive circuit and no cavity. Since then:

- v0_1 to v0_6: the Boltzmann equation solved in the loop (v0_3); the 5p rung, the 248-nm optics and Turnbull's inverse bremsstrahlung for an EM wave (v0_4); the measured 5s $\rightarrow$ 5p excitation (v0_5); the reheat test (v0_6);
- 1-D versions on Fluid1D (1Dv0 to 1Dv0_3);
- copies for ArF, ArF under an electron beam, and pure krypton.

## 3 The four side by side

| | 1. ODE | 2. phantom | 3. SPECK | 4. LIGHT |
|---|---|---|---|---|
| when | before June 2026, undated | before June 2026 to 3 August | 3 August to 18 September (REBUILD 0.1.0 to 0.1.40) | 25 September to now (v0 to v0_6) |
| where | `KrF_and_LoKI/KrF Code/og-pyfiles/`, frozen | `KrF_and_LoKI/KrF Code/ factored-pyfiles/` and the root GUIs | `SPECK/REBUILD/`, `SPECK/PIC_MCC_3D/` | `LIGHT/`, with copies `LIGHT_ArF/`, `LIGHT_ ArF_ebeam/`, `LIGHT_ Kr/` |
| model | 0-D; 11 and 13 variables | 0-D, 26 species; 1-D and 2-D $CO_2$ extensions | 0-D or a 3-D grid of independent cells; 35 species, 174 reaction rows in KrF mode | 0-D $Kr/F_2$; 24 rows (v0), 37 (v0_1); 1-D versions on Fluid1D |
| electron rates | Maxwellian averages | LoKI-MC tables from v26, at an electron temperature tuned to Chow | LoKI-MC tables for three fixed fills, or rates measured by PIC_MCC_3D | a Maxwellian with a kinetic tail, or the Boltzmann equation solved in the loop (v0_3 on) |
| cost to run | not recorded | seconds per 0-D run | 5 to 10 s per 0-D point; hours for a $2 \times 2$ grid | under 1 s (v0); minutes with the Boltzmann equation in the loop |
| Python lines, tests excluded | 505 and 220 | a 9,798-line file, then 12,475 (engine) and 10,677 (GUI) | about 14,000 (REBUILD) and 5,500 (PIC_MCC_3D) | about 2,200 (v0), growing to 7,400 (v0_6) |
| scored against | aimed at Chow 1978; no score recorded | Chow 5/6, Ong 5/5, Chen 5/5, all found circular | Klimek gain $\times 0.64$, Tahoma $\times 0.58$, Electra ArF $\times 0.75$; held-out KrF decays $\times 1.37$ | held-out KrF decays $\times 0.95$; Tahoma gain $\times 0.93$ |
| how it ended | grew into phantom; kept as the parity reference | audit, bugs, benchmark path banned 13 July, ruled flawed 3 August | kept: the only model here for buffers, real drives, a cavity or a grid | the current 0-D reference |

Table 1: The four generations side by side. The line counts were made for this overview on 20 October 2026; every other entry is quoted from the files of Section 6.

## 4 The thread through all four: does reheating pay?

Each generation answered the reheating question, and each answer was smaller than the one before.

- **phantom.** A $CO_2$ pulse raised the efficiency by about 4 %. The audit found that number an upper bound resting on an inversion the code itself forbids elsewhere, which a 2-D run collapses to +0.06 % (`docs/PHANTOM_AUDIT.md`); on the production model the microwave gives +0.000 % (PLAN entry (73)).

- **SPECK.** "A robust observable, a real but marginal channel" (`SPECK/PAPER/`). The fluorescence rises in all eight benchmarks, but no benchmark's efficiency moves by more than 0.006 percentage points (REBUILD 0.1.20 to 0.1.28). The independent 1-D model (`SPECKLE/`) shows the same fluorescence rise, and the 3-D grid (`SPECKLE3D/`) reproduces the small afterglow gain.
- **LIGHT v0_6.** At LIGHT's defaults the reheat lowers the efficiency everywhere. It pays, by up to 0.14 point, only if one rate nobody has measured, $Kr_2^* + F \rightarrow KrF(B)$, is above about $2 \times 10^{-12}\,cm^3/s$ (under 1 % of Kannari's estimate), with a $3\,MW/cm^2$ wave and a source at least half as efficient as the pump. Under the two-temperature closure, or at 2 % $F_2$, it never pays (`LIGHT/v0_6/results/reheat.md`).

# 5 Where everything lives

| directory | what it holds |
|---|---|
| **LIGHT, the current generation** | |
| `LIGHT/` | the 0-D model, v0 to v0_6, and the 1-D models 1Dv0 to 1Dv0_3 on Fluid1D; accuracy overviews and system maps in `LIGHT/docs/` |
| `LIGHT_ArF/`, `LIGHT_ArF_ebeam/`, `LIGHT_Kr/` | LIGHT made to model ArF, ArF under an electron beam, and pure krypton |
| `ML/` | LIGHT's channels fitted to the measurements (least squares, MCMC, a neural network), with laser data kept for scoring only |
| `comparisons/` | every model scored on the same measurements; the verdict for 0-D and 1-D is at the top of its README |
| `KrF_model_notes_with_Gary.rtf` | the notes LIGHT was built from |
| **SPECK** | |
| `SPECK/REBUILD/` | the chemistry engine: e-beam and discharge drives, cavity, 3-D ASE, the gain anchors |
| `SPECK/PIC_MCC_3D/` | the 3-D electrostatic particle-in-cell code whose measured rates feed REBUILD |
| `SPECK/PAPER/`, `SPECK/COMMERCIAL_AUDIT.md` | the microwave-reheat article draft and talk; what stops the results from commercial-grade confidence |
| `HIBACHI/`, `hibachi_out/` | the e-beam driver chain (pulsed power, diode, foil) feeding REBUILD, and its plots |
| `SPECKLE/`, `SPECKLE3D/` | the reheat question checked on Fluid1D (1-D) and on REBUILD's 3-D grid |
| **phantom and the ODE** | |
| `KrF_and_LoKI/` | the frozen originals (`og-pyfiles/`), the `phantom11` engine, the `unified` GUI back end, the 1-D and 2-D PIC demonstrators, spectre, and phantom's version history in `docs/` |
| `GUI_v2.py`, `RUN_HYPERLASER_GUI.py` | the phantom-era GUIs (v2 and v1) |
| `Argon_UI/`, `ARPAE/` | an argon-laser prototype model, and a 3-D electromagnetic PIC with a horn antenna, both from late July |
| `PLAN_physical_validity.txt` | the phantom era's audit log, entries (1) to (74), under the policy |
| `README_LIMITS.md` | what the 1-D PIC model can and cannot do |
| `old_files/`, `old_guis/` | quarantined code and retired launchers; nothing there is working code |
| **Outside code and references** | |
| `LoKI-MC/` | the LoKI-MC electron-kinetics solver (C++, GPL) and its inputs |
| `nathaniel/1DModelKrF/` | Nathaniel's Fluid1D, the 1-D discharge model (AGPL), used read-only |
| `osiris/` | the OSIRIS particle-in-cell code, never built here |
| `docs/`, `Recent_KrF_Papers/`, `new_articles/`, `PIC_papers/`, `yu1990.pdf` | the literature, Chow 1978 (`docs/KrF_Chow.pdf`) first |
| `presentation/`, `IMAGES/`, `air_exp/` | the 1 July update talk and derivation, PIC animations, and a spark-gap side study |

Table 2: The top-level directories, grouped by the generation they belong to.

## 6 Where the record is

The git history is detailed only to 31 July 2026 (PLAN entry (74)). After that a few large commits carry the work, and each sub-project's own `CHANGELOG.md`, Known-bad list first, is the record. To read the history in its own words:

- the ODE and phantom: `KrF_and_LoKI/docs/VERSION_HISTORY.md`, `KrF_and_LoKI/docs/CHANGELOG.md`, `docs/PHANTOM_AUDIT.md`, `KrF_and_LoKI/docs/two-region-eedf.md`, `PLAN_physical_validity.txt`, `old_files/README.md`;
- SPECK: `SPECK/ORESTES_CLASS_REBUILD.txt` (why and how the rebuild), `SPECK/README.md`, `SPECK/REBUILD/CHANGELOG.md`, `SPECK/PIC_MCC_3D/CHANGELOG.md`;
- LIGHT: `KrF_model_notes_with_Gary.rtf`, `LIGHT/README.md`, `LIGHT/CHANGELOG.md`, `comparisons/README.md`;
- the root: `README.md` and `CHANGELOG.md`.

This overview was written from those files on 20 October 2026. No model was run and no code was changed to make it; its numbers are quoted from the files, not measured again, except the line counts in Table 1.